Universidad de La Laguna

Introduction to Design Patterns

Academic Year 2025–2026

Introduction to Design Patterns

Link to code examples in these slides
Link to all code examples in this course

Don't forget to update the repos!

Contact us!

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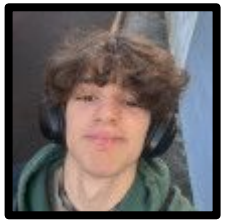

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- Why should you be interested?
- What is a design pattern?
- Some background information
- Why there a
- Classification of design patterns

Why should you be interested?

Why should you be interested?

PAI 2526 / Práctica ?? - ??? / 2025-2026 Cuestionario ??. Introduction to Design Patterns

2025-2026 Cuestionario ??. Introduction to Design Patterns

Por hacer: Recibir una calificación

Abre: ???

Cierra: ???

Why should you be interested?

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2025-2026 Cuestionario ??. Introduction to Design Patterns

Por hacer: Recibir una calificación

Abre: ???

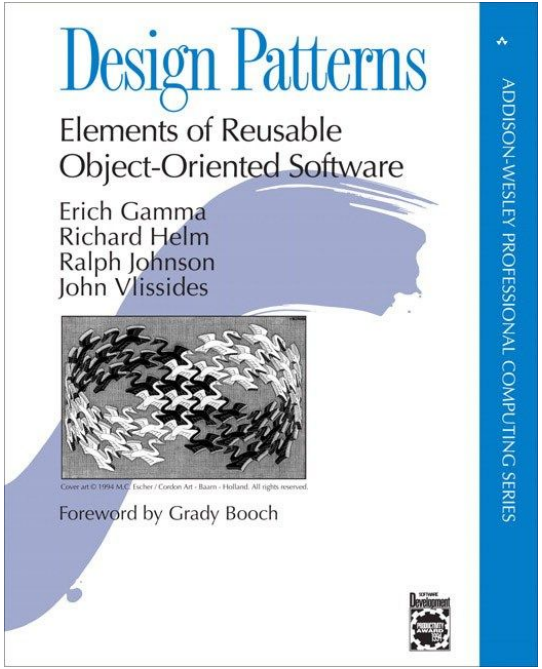
Cierra: ???

That’s a good reason, but there are better ones…

Some background information

Design Patterns

Gang of Four (GOF)



[More info here!]

What is a design pattern?

“Design patterns are typical solutions to common problems in software design.”

Source: Refactoring Guru

Why are they important?

- Easy to engage in a new project

Why are they important?

- Easy to engage in a new project
- Universal understanding

Why are they important?

- Easy to engage in a new project
- Universal understanding
- Documented methodology

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- Easy to engage in a new project
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- Documented methodology
- Industrial standards

Why are they important?

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- Industrial standards

More in detail

How to use them

It is NOT about copying and pasting…

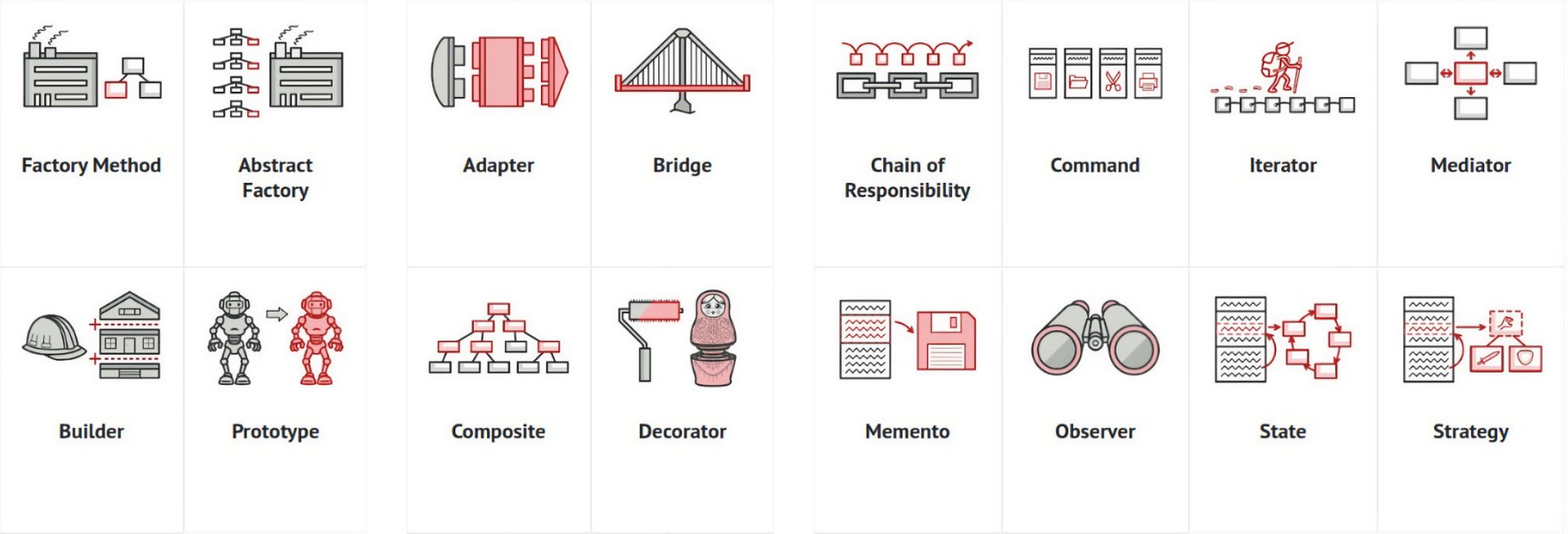
How to use them

It is NOT about copying and pasting…

…rather **computational thinking!**

There are a lot of them…

There are a lot of them…



There are a lot of them…

We will only cover the easiest to understand patterns!

Classification

Classification

- Creational

Classification

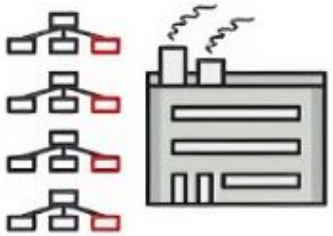
- Creational
- Structural

Classification

- Creational
- Structural
- Behavioural

Creational

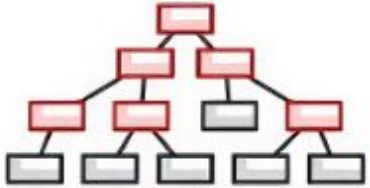
Defines how objects are created.



More about creational patterns…

Structural

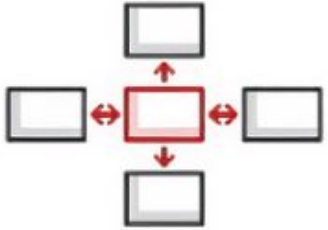
Defines how objects relate to each other.



More about structural patterns…

Behavioral

Define how objects communicate with each other.



More about behavioural patterns...

Let’s dive into a few of them…

- Deep detail.
- Only a few of them.

Builder pattern

Intent

...

image of guru

Problem

...

image of guru

Solution

...

image guru

Real life example

...

image

Code example

...

image

source of code

Facade pattern

Intent

...

image of guru

Problem

...

image of guru

Solution

...

image guru

Real life example

...

image

Code example

...

image

source of code

Singleton pattern

Intent

...

image of guru

Problem

...

image of guru

Solution

...

image guru

Real life example

...

image

Code example

...

image

source of code

Adapter pattern

Intent

...

image of guru

Problem

...

image of guru

Solution

...

image guru

Real life example

...

image

Code example

...

image

source of code

Mediator pattern

Intent

...

image of guru

Problem

...

image of guru

Solution

...

image guru

Real life example

...

image

Code example

...

image

source of code

Generic title

Generic text

Generic title

Generic text

- Generic bullet point list
- …

Generic title

- Generic bullet point list
- …

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Which pattern is the best?

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The one that best solves your problem!

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Any questions?

References

- C# Design Patterns (dofactory)
- Refactoring Guru
- GeeksForGeeks
- Design Patterns by the GoF (Amazon)